Daniel **McSheehy**

SENIOR SOFTWARE ENGINEER

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"You are the universe experiencing itself"

Summary

Senior Software Engineer with 7+ years of experience building distributed backend systems, autonomous AI agent platforms, high-throughput telemetry pipelines, and web applications.

Work Experience

Formant

San Francisco, California (Remote)

SENIOR SOFTWARE ENGINEER (PROMOTED FEB. 2023) — *Oct. 2021 - Present*

- Built the persistent-agent autonomy platform and sandboxed code execution runtime (bwrap TypeScript/Python, recursive LLM sub-calls, durable event wakes, and graph-linked pgvector memory) for 24/7 fleet operations and native robot onboarding.
- Architected the knowledge graph and evaluation stack: content-addressed document sets in MinIO, LightRAG hybrid semantic search, polymorphic LLM/ground-truth judges, and multi-turn orchestration benchmarks.
- Designed the AI task-flow engine and BullMQ execution layer; cut LLM token costs and latency up to 80% via compact context references and prompt caching, and resolved 5+ production heap/queue memory leaks.
- Built the self-serve telemetry analytics engine across Snowflake and ClickHouse with ECharts modules, row-level security via tag-matching SQL UDFs, gap-filling time series, and LRU/Redis caching.
- Authored the platform's RBAC and resource-policy system across PostgreSQL, NestJS, and React, and built the organization AI usage and billing pipeline.

effx

San Francisco, California

SOFTWARE ENGINEER — *June. 2019 - Oct. 2021*

- Designed, built, and maintained interstellar, a distributed jobs scheduler running thousands of jobs a minute.
- Built a large portion of the company's front-end React web app and Terraform AWS IAM integration modules.
- Built 6+ Go microservices operating on MySQL and Elasticsearch, instrumented with Datadog SLOs.

Hyperledger

Remote

SOFTWARE ENGINEERING INTERN — *May. 2018 - December. 2018*

- Built and became lead maintainer of an open-source Chrome extension that connects to Hyperledger Fabric networks.

RankScience (Y-Combinator W17)

San Francisco, California

SOFTWARE ENGINEERING INTERN — *May. 2017 - Dec. 2017*

- Built production Node.js backend services backed by MongoDB and improved React frontend state management and test coverage.

Skills

LANGUAGES / FRAMEWORKS / INFRASTRUCTURE

- **TypeScript / JavaScript (Advanced):** Node.js, NestJS, React, Redux, Zustand, Vite, ECharts, GraphQL
- **Systems & Backend:** Go, Rust, Python, gRPC, Microservices, BullMQ, WebSockets
- **AI & Agents:** LLM Orchestration, Mastra, LightRAG, Knowledge Graphs, MCP, Sandboxed Code Mode, Evaluation Harnesses
- **Data & Storage:** PostgreSQL (pgvector), ClickHouse, Snowflake, TimescaleDB, Redis, MySQL, Elasticsearch, MinIO / S3
- **Infrastructure:** AWS, Docker, Kubernetes, Helm, Terraform, Vault, Datadog, GitHub Actions

Honors & Awards

INTERNATIONAL

2018 **1st Place**, Accenture's Blockchain for Good Hackathon - Team: Decentradelized (Farming supply chain) — *Basel, Switzerland*

DOMESTIC

2018 **1st Place**, Proof of Hack (Blockchain Hackathon) - Team: DataHash (decentralized data transfer) — *Dallas, Texas*

Education

University of Texas at Dallas

Dallas, Texas

B.S. IN ELECTRICAL ENGINEERING — *May 2019*

- Gained valuable knowledge about electrical systems, physics, and solving complex engineering problems.